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Beyond 1 Million Nodes—Crowdsourced Video CDN: Architecture, Technology, and Economy

Lifeng Sun, *Member, IEEE,* Ming Ma, Wen Hu, Haitian Pang, and Zhi Wang, *Member, IEEE.*

***Abstract*—To deliver the ever growing video traffic with the Quality of Experience guarantee, deploying the infrastructure of Content Delivery Network closer and closer to users has been a noticeable trend. And this trend becomes more promising when recently CDNs providers start utilizing the storage and network resources of the edge network devices for content delivery, i.e., crowdsourced CDN. In this article, we envision the crowdsourced CDN and propose a set of practical strategies to guide the implementation of such new paradigm. Firstly, we perform a large-scale measurement to explore the geo-content popularity. Benefiting from the observations, we design the key solutions for the system operation, including regional content popularity prediction, region partition and the collaborative content replication. Moreover, we consider the economic issues and propose an auction mechanism to cope with the resource competition among multiple content providers. We also demonstrate the performance gain of our design with data-driven simulation.**

***Index Terms*—Content delivery network, Crowdsourced CDN, Edge caching.**

I. Introduction

FUELED by the growing prevalence of Ultra High Definition videos and Internet-capable portable devices [1], [2], the soar of video traffic extremely consumes the network bandwidth resource. To alleviate the backbone network traffic and improve the service quality experienced by users (e.g., reduce the service delay), pushing video services and infrastructure closer to users have become a common solution and gained interests both from the academic and industrial fields [7], [14]. The traditional approach for Content Delivery Network (CDN) to provide "close-to-user" video service resorts to deploy more infrastructure or rent more Cloud resources [3], [16], e.g., Akamai, one of the largest CDN providers in the world, currently owns more than 216K servers in over 120 countries and 1.5K ISPs around the world. But it is expensive for CDNs to deploy this plenty of edge servers. While in recent two years, the emergence of crowdsourced video CDNs [5], [9], [13], which utilize abundant of existing edge devices to assist the video delivery, mitigates this limitation, such as femtocell [8], set-top boxes [11], smartrouters [13]. The edge devices (referred to as content hotspots) are encouraged to contribute their available storage to cache contents and upload bandwidth to distribute contents to end users by cash rewarding. And hence the crowdsourced CDN can provide the closest video access to end users (also implying a low-latency and high-bandwidth service) and bring lower expenses to Content Providers (CP), e.g., iQiyi, Youku [13] and Thunder [5], three famous video providers in China, has deployed hundreds of thousands of content hotspots at their users' homes for their video delivery, expecting to turn a large fraction of its users to such edge devices. With this rapidly growing speed of the edge resource deployment, we should consider the challenges brought from the system utilization in the case of controlling millions of the content hotspots even in one city.

To guide the implementation of crowdsourced CDN, it is intriguing to investigate the problems about the caching efficiency and economic issues. Firstly, to design the fundamental strategies for the edge caching, we should solve the hotspot organization problem and the collaborative content replication problem (i.e., which contents should be replicated to which content hotspots from which content source to collaboratively serve users). It is challenging to optimize system performance over millions of content hotspots which disperse across multiple regions due to the scalability issue. Secondly, in the market, when multiple CPs compete for the fixed edge resources of one crowdsourced CDN provider, the selfishness of each party may cause resource waste. But the resource allocation on each hotspot is coupled with each other, making the collaborative caching and pricing problem complex and difficult to solve.

Previous studies on the strategies design of the crowdsourced CDN can be divided into two categories. The first is *global scheduling solution* [5], [13], which determines the number of content replicas based on the globally content popularity prediction and ignores the difference between the user interests in different locations, making such global scheduling solution inefficient [12]. The second one is *mathematical based solution* [6], [8], which mainly explores the mathematical analysis for the content placement problems by maximizing the request ratio which could be served by the edge content hotspots. Although their algorithms capture the local content popularity to some extent, these are impractical (e.g., requiring high running time) to control the massively distributed content hotspots.

To this end, we carry out a large-scale measurement study to learn the regional content request patterns and the possibility of using today's edge-network devices for content delivery. With the valuable insights from our measurement studies, we emphasize the geological heterogeneity on both distributions of end users and content hotspots and propose a set of data-

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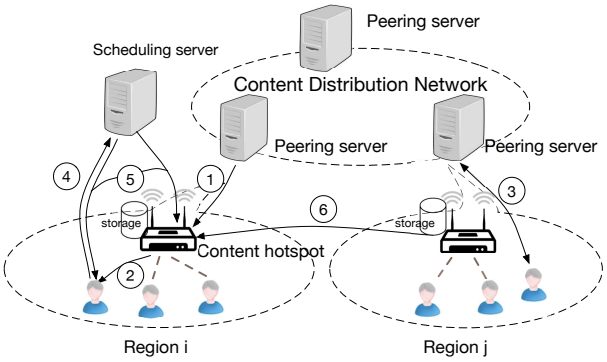

Fig. 1: The architecture of crowdsourced CDN.

driven strategies to guide the implementation of crowdsourced CDN. In particular, we formulate the user interest transition as a Markov Decision Process (MDP) to predict the regional content popularity. And we discover the trade-off in crowdsourced CDN. If content hotspots are deployed close to users, they can perceive high Quality of Experience (QoE) [15]; but replicating contents to hotspots also results in higher replication costs. Hence we propose a user-aware region partition algorithm and the content hotspot management could be organized by the generated regions, to achieve a joint objective of improving users QoE and reducing the replication cost. We further design a cross-region cache cooperation strategy for the content replication, which collaboratively replicates contents from the centralized server or other regions. Especially, we take both the local content preference and the similarity among neighbouring regions into account.

For the economic issue, we study the pricing mechanism and resource allocation mechanism to look into the resource competition among CPs. When multiple CPs compete for the edge hotspot resources to deliver video contents, it is essential to design an efficient resource allocation mechanism. We propose a Sequential Auction Mechanism (SAM) for resource allocation among CPs which can guarantee Individual Rational, Incentive Compatible, and at least $\frac{1}{2}$ optimal performance.

The rest of this article is organized as follows. First, we present the overall system architecture and workflows for the crowdsourced CDN. Next, we propose a set of key schemes to guide the system implementation, including the content popularity prediction, regional resource organization, and content replication with cache collaboration. Then we describe our resource allocation mechanism when multiple content providers (CPs) compete for the edge resources. Finally, we summarize this article.

II. Crowdsourced CDN Architecture

Crowdsourced Video CDN enables edge content hotspots to share and cooperate their storage/bandwidth resources. And users access the content hotspots through edge backhaul link. As shown in Fig. 1, such system comprises three complementary components, including content hotspots, CDN servers, and centralized scheduling servers. Each of them is explained as follows:

▷ Content hotspots are the dedicated edge devices, with the intelligent delivery software and the built-in storage, which can proactively cache (step ①) and distribute video (we changeably use content and video) to users (step ②).

▷ CDN servers own all of the contents and take two responsibilities in the system. The first is keeping publish latest videos to the content hotspots (step ①), and the second is as general servers for the video downloading of users if none available content hotspots could serve the user (step ③).

▷ Centralized scheduling servers are the key component, which collects and learns the global system information (step ④) and decides the primary scheduling strategy (step ⑤), e.g., which videos should be replicated to which content hotspots in step ① and which users should be served by which content hotspots in step ②. In the real-world system, the content hotspots periodically query the scheduling server about prefetching and caching which contents in the next timeslot.

In this article, we propose three key strategies for the Crowdsourced Video CDN introduced as follows:

- Content popularity prediction function: It models the diverse regional content preference based on the real traces collected from real-world systems.
- Region partition function: It is used for the content hotspot organization, and we strike a balance between the extra content placement cost (incurred by the proactive caching) and the improved users' QoE through the region partition process.
- Content replication function: Based on the regional content popularity and the area partition function, we conduct the content replication by taking the time-varying server load and the content similarity among neighbouring regions into account (step ②).

Next, we detail these strategies.

III. Key System Technologies

In this section, we conduct a large-scale measurement study to explore the geological difference on the content popularity and content hotspot distribution. Based on the observation, we present the key technological strategies to solve the challenge problems of content hotspot management and content placement.

A. Content Popularity Study with Regional Preference

Content popularity prediction is desired to accurately capture the user preference from a large scale geo-location distributed areas. To study how users watch videos in today's mobile Internet, we use a data-driven approach based on the video traces from iQiyi, one of the largest online video providers in China. The traces were collected from 1.8M users in Beijing City during 2 weeks of May 2015, containing 59M sessions from 0.4M videos. Each video request trace records the user identifier, the title of the requested video, the timestamp, and the location where the user issues the request.

Based on the dataset, we divide the area (i.e., Beijing) into $N \times N$ cells, and design top-K content similarity index to reflect the similarity of top-K popular contents between one cell and its adjacent cells. Specifically, we let $V_i(K)$ denote

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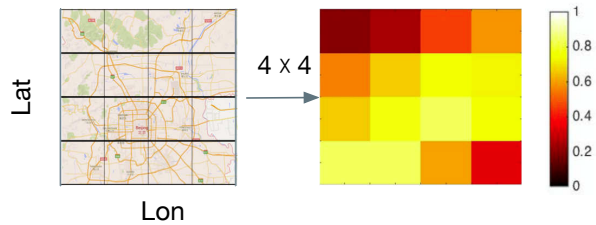
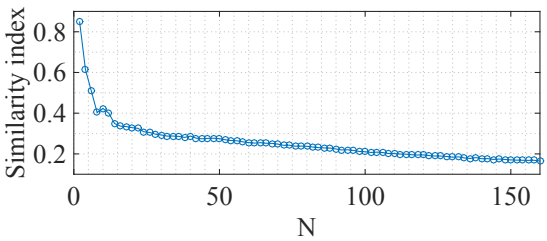


(a) Heatmap of content similarity index, $N = 4$.

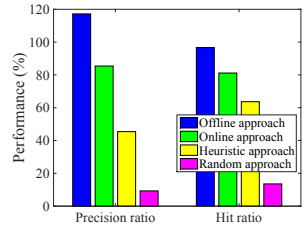


(b) Average content similarity index versus N.

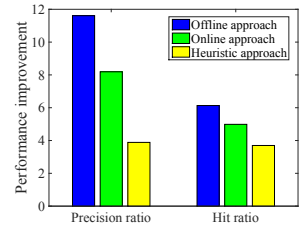
Fig. 2: Top-K content similarity index for regional cells, $K = 10$.

the set of top-K contents in cell i and let $V_i'(K)$ denote the set of top-K contents in i's adjacent cells. Then based on Jaccard similarity coefficient, our top-K content similarity index can be calculated as $\frac{|V_i(K) \cap V_i'(K)|}{|V_i(K) \cup V_i'(K)|}$. E.g., Fig. 2(a) illustrates the geo-distribution of content similarity index with $K = 10$ and $N = 4$. Furthermore, in Fig. 2(b), we plot the average similarity index of all cells versus N in the partition. We observe that the average similarity index decreases when N increases (i.e., cell is partitioned smaller). This indicates that when the cell size is smaller, users in different cells tend to have dramatically different content preferences. Hence using the regional/local content popularity can potentially serve the heterogeneous requests more efficiently than centralized content delivery paradigms.

To address the challenge of content popularity prediction with such "small population" in the crowdsourced CDN [12], we employ the content semantic information and users' watching behavior to guide the design of prediction strategies. For example, we utilize the semantic information of TV series (such as the content title, the content genre, the content introduction), to understand users' affinity to the contents and users' video transition behavior. By studying the users' TV series watching pattern, we observe users tend to watch the consecutive episodes in the same TV series, e.g., users are prone to either keep watching the current episode or the next three episodes, with a probability of 35% and 47% respectively. Based on this insights, we model users transition behavior among series s a MDP. In particular, we adopt the reinforcement learning algorithm to design an online strategy for such content popularity prediction, which is more practical in real system, and we choose three other algorithms as baselines: i) an offline approach, which obtains the upper bound based on the availability of users' future behavior; ii) a heuristic approach, which assumes users are prone to either keep watching the current episode or the next three episodes; iii) a random approach, which assumes users randomly watch

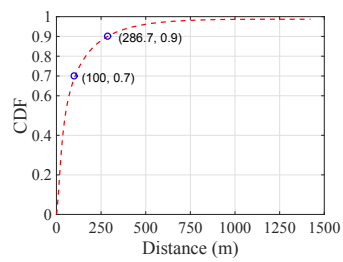


(a) Precision and hit ratio with different approaches.

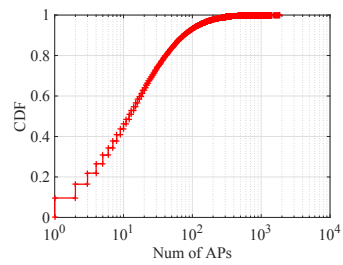


(b) Improvement compared with random approach.

Fig. 3: Performance comparison in terms of precision and hit ratio.



(a) The cumulative request ratio vs. the distance between the requests and their nearest APs.



(b) The distribution of the number of APs located within $100m$ to a request.

Fig. 4: Resource availability.

a following video. As shown in Fig. 3, we observe our online approach can achieve about 80% precision accuracy, which is 9 (resp. 1.8) times better than the random (resp. heuristic) approach. As for the hit ratio, the online approach is about 6 (resp. 1.3) times better than the random (resp. heuristic) approach.

Then we consider how to appropriately partition the area for the regional content popularity prediction, rather than using the naive partition method in Fig. 2.

B. Partition for Edge-network Areas

Firstly, we study the geographical resource availability based on a 1M Wi-Fi Access Point (AP) dataset in Beijing City. We correlate the APs with the iQiyi video requests by their location and let these APs be the content hotspots in our crowdsourced CDN, to measure how well the requests can be served by edge content hotspots nearby. In Fig. 4(a), we plot the cumulative request ratio versus the distance between the location where the request is issued and the nearest AP. We observe that for over 70% (resp. 90%) of the requests, users can find at least 1 Wi-Fi AP within 100m (resp. 286.6m), indicating that using Wi-Fi APs as edge-network content hotspots is promising. Furthermore, the curve in Fig. 4(b) illustrates the CDF of the number of APs within 100m around a user request, and this heterogeneous resource distribution indicates that we should strategical partition the area to organize the massively distributed content hotspots for the content replication.

In the crowdsourced CDN, on the one hand, users want to download contents from content hotspots as close as possible to achieve better QoE. On the other hand, the content replication cost increases when users only fetch videos from too

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close content hotspots, i.e., in the most extreme case, every content hotspot has to cache all possible contents to serve its nearby users. Hence it is reasonable to find an appropriate area partition solution and assume that all video requests issued from a partitioned sub-area will be served by the content hotspots in this sub-area.

Let $P = \{(\mathcal{R}_1, \mathcal{U}_1), (\mathcal{R}_2, \mathcal{U}_2), ..., (\mathcal{R}_M, \mathcal{U}_M)\}$ denote a partition, where the area is partitioned into M cells $\{\mathcal{R}_1, \mathcal{R}_2, ..., \mathcal{R}_M\}$, and users in the area are accordingly partitioned into $\{\mathcal{U}_1, \mathcal{U}_2, ..., \mathcal{U}_M\}$. We design a utility function to balance the tradeoff of users' QoE and content placement cost for a particular area partition P [10]:

$$g(P) = \alpha \cdot \sum_{i=1}^{M} \sum_{u \in \mathcal{U}_i} log(B_{u,n_i}) - \beta \cdot \sum_{i=1}^{M} \sum_{u \in \mathcal{U}_i} D_{u,n_i} - \gamma \cdot \sum_{i=1}^{M} \|\mathcal{V}_i\|, \tag{1}$$

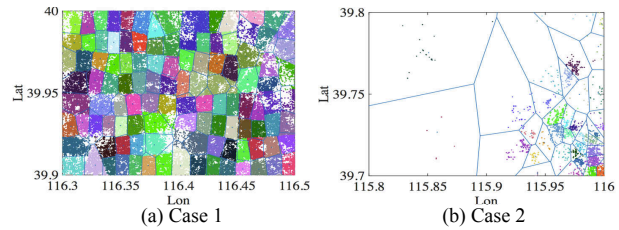
where n_i is a virtual hotspot which locates at the center of cell i, to represent all hotspots in cell i, B_{u,n_i} denotes the bandwidth perceived by user u downloading content from n_i, and D_{u,n_i} denotes the distance between user u and content hotspot n_i. Motivate by [4], $\alpha \cdot \sum_{i=1}^{M} \sum_{u \in \mathcal{U}_i} log(B_{u,n_i}) - \beta \cdot \sum_{i=1}^{M} \sum_{u \in \mathcal{U}_i} D_{u,n_i}$ denotes the users' QoE of the system. $\mathcal{V}_i$ denotes the set of contents requested by users in cell i, and hence $\sum_{i=1}^{M} \|\mathcal{V}_i\|$ is the content placement cost of the system. The rationale of $g(P)$ is that we are trying to jointly maximize the QoE and minimize the replication cost, and the weight of the two objectives can be adjusted by varying the parameter α, β and γ. Thus the area partition problem can be formulated as: $\max_P g(P)$.

We design Voronoi-like solution to solve it greedily. At every iteration, the greedy algorithm selects an area with the highest marginal utility, i.e., determine to partition which area into sub-areas could get the highest utility increase. Fig. 5 presents the partition results of 2 cases. Each sample indicates a location where a request is issued, and the sample color represents the sub-area to which the sample belongs. We observe that our partition result is similar to $N \times N$ grid-based partition under uniform geo-distribution of requests (Case 1), and our edge-network partition is highly adaptive to the geo-distribution of the user requests (Case 2). We evaluate the improvement of our Voronoi-like partition algorithm by comparing it with the $N \times N$ grid-based partition in terms of different metrics, and we observe our design can improve the users' QoE by 44% in Case 2 by sacrificing smaller extra replication cost (18%) to the system. In general, our design improves the utility up to 25%, which considers QoE and cost simultaneously.

C. Collaborative Content Replication

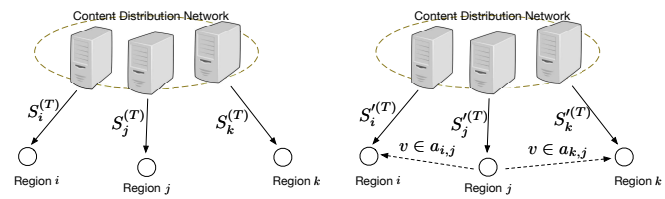
In this part, we present how to efficiently replicating contents to these edge-network content hotspots. Based on the regions generated from the content hotspots management algorithm in Sec III-B, we conduct explore the design space of content replication as follows.

First, to improve the replication efficiency and reduce the redundant traffic incurred by replication for the same content, we propose to utilize the similarities among regions



(a) Case 1 (b) Case 2

Fig. 5: The results of Voronoi-like partition.



(a) Regional popularity based content replication. (b) Collaborative content replication.

Fig. 6: Comparison between non-collaborative and collaborative content replication.

for collaborative replication, i.e., the original CDN servers do not need to push all the contents requested by users in one region and a content stored in one region can be replicated to other regions wherein this content is not available, to satisfy the corresponding user requests. As illustrated in Fig. 6(a), the non-collaborative content replication is pushing content only based on the regional content popularity (e.g., $S_j^{(T)}$). Compared with such non-collaborative method, we propose to design a collaborative content replication strategy. As illustrated in Fig. 6(b), we strategically decide whether content v should be pushed to region j by jointly considering the offloading effect for region j and its neighbouring regions (e.g., region i and region k).

Next, to make the best use of limited available edge resources (e.g., the bandwidth and storage), it is essential to prioritize the order of content replication. We build a content offload index $G_{i,k}^{(T)}$, which represents the offload efficiency index by replicating content v_k from the original CDN servers to region i at timeslot T. In the case of collaborative replication, $G_{i,k}^{(T)}$ is affected by many factors, including the intensity of local requests, the intensity of remote requests, the number of remote regions, the distance between the remote regions and region i. We define $G_{i,k}^{(T)}$ as below.

$$G_{i,k}^{(T)} = \mathcal{E}_{i,k}^{(T)} + \sum_{j=1}^{M} \frac{\mathcal{E}_{j,k}^{(T)}}{\mathcal{H}(d_{j,i})}, \tag{2}$$

where the $\mathcal{E}_{i,k}^{(T)}$ and $\mathcal{E}_{j,k}^{(T)}$ is the requests volume for content v_k from region i and region j at timeslot T, respectively. $\mathcal{H}(d_{j,i})$ is the weight function with respect to the distance $d_{j,i}$.

Based on the offload efficiency index, our proposed replication works as follows. Following reports from all the edge-network content hotspots, the scheduling server calculates how much traffic can be offloaded if a content is replicated to a particular content hotspot, according to the predicted requests

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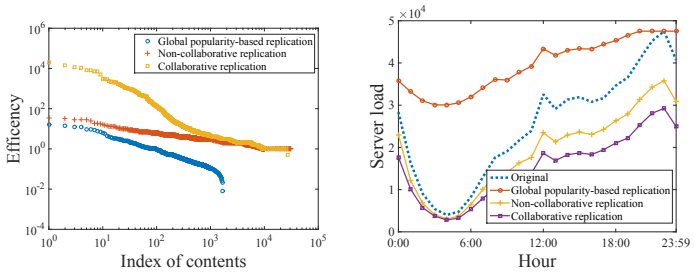
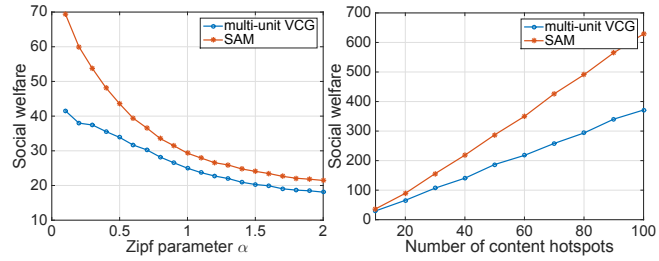


(a) Efficiency of content replication. (b) Server load over time.

Fig. 7: The performance of content replication algorithm.



(a) The change of the popularity of CPs. (b) The change of the number of edge content hotspots.

Fig. 8: Social welfare under different metrics.

of the content from all hotspots. For each content-region pair, if the nearby regions have stored the content and have enough resource to conduct the inter-region replication, the inter-replication will be conducted. Otherwise, the content is replicated from the server. The replication continues until all content have been replicated or the server load reaches the observed peak traffic.

In Fig. 7(a), we compare the efficiency of globally popularity-based replication [5], non-collaborative replication and our proposed collaborative content replication strategy. We observe that our content replication algorithm which considers the content diversity in different sites and takes advantage of inter-regions collaboration highly overpasses the global popularity based content replication strategy and non-collaborative strategy. The reason is that the globally popular content items may be not the hot content items at every cell, whereas geo-preference-aware replication strategies (e.g., non-collaborative and collaborative methods) pushes content items in the site grain which means it will actually serve many requests in the corresponding sites. In addition, our proposed collaborative replication further improves the offload efficiency of the replicated content items by utilizing the inter-region replication. The server load reduction after adopting our proposed collaborative content replication is shown in Fig. 7(b). We observe that, compared with the global popularity based strategy, the non-collaborative algorithm can reduce server load up to 27%. Furthermore, when our design takes the similarity among nearby regions into account to collaboratively conduct the content replication, the server load can be further reduced to 38%.

IV. Auction-based Resource Competition

Considering the fact that, in the market, there are multiple CPs to compete the fixed content hotspot resources in the crowdsourced CDN system. In the above Sec III, we only discuss the caching technologies for a single CP. But the problem becomes more complex when we consider both collaborative caching scheme and cache resource allocation among multiple CPs.

For the economic point of view, the whole crowdsourced CDN ecosystem is made up of three parties, including the content hotspot owners, the crowdsourced CDN providers who crowdsource these content hotspots from their owners, and the CPs who leases the cache space from crowdsourced CDN providers. Then we study the following questions systematically: i) How should the crowdsourced CDN allocate the cache spaces over the CPs? And ii) How should crowdsourced CDN provider design his selling scheme to fit this network architecture and improve the overall performance of the network? From crowdsourced CDN providers' perspective, maximizing the profits from all of the CPs is the target; From CPs' perspective, the most crucial benefit of utilizing more content hotspots is to get more reduction on the transmission cost (i.e., the shortened the delivery latency), but much payment is unexpected. As the crowdsourced CDN provider and CPs will pursue his benefit, there may be resource waste in the system. Hence the essential problem is how to design resource allocation mechanisms to achieve the social optimality.

To solve this problem, we devise Sequential Auction Mechanism (SAM). In the SAM, the auctioneer will perform an auction for each cache space in each content hotspot sequentially. For each auction, the auctioneer will ask each CP to submit his bid, which is the price he is willing to pay to get the caching space. After receiving the bids, the auctioneer will allocate this cache space to the bidder who offers the highest price and the corresponding bidder will place a content to this cache space. After each caching decision, the CPs will recalculate the transmission cost, as the newly cached content will affect where the content is fetched. Each CP has a prediction of content popularity which is his private information. This mechanism guarantees that each CP should submit the price of his real transmission cost reduction to get the optimal payoff. This mechanism also guarantees the decouple of the problem of pricing and caching, as the caching decision is determined sequentially. Our algorithm preserves the economic property of Individual Rational, Incentive Compatible and at least $\frac{1}{2}$ optimal. And it solves the problem in polynomial time, which is $\mathcal{O}(S \cdot \log(N))$, where S is the total number of cache space and N is the number CPs.

Then we evaluate the solution performance. We utilize the classical Vickrey-Clarke-Groves (VCG) auction mechanism as the baseline. Since the caches on one content hotspot are homogeneous, the allocation of the caches on one content hotspots can be solved by a multi-unit VCG auction. Then we run the multi-unit VCG auction on each content hotspot to solve the collaborative caching problem. We adopt the Zipf distribution as the popularity distribution over CPs, and α denotes the CPs' difference in popularity. Fig. 8(a) denotes the

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social welfare under the different α. We observe that when α is relatively small, i.e., the CPs' popularity is similar, SAM outperforms multi-unit VCG by 75%. Fig. 8(b) shows the social welfare under different number of edge content hotspots. We observe that SAM show more advantage when the node number is large, as SAM considers the caching collaboration.

V. Conclusions

Crowdsourced CDN promotes video streaming services by enabling edge devices to provide services to end users. In this article, we propose key strategies for system implementation and economic issue, which is applicable for the common scenario that millions of content hotspots need to be scheduled and multiple CPs compete for the fixed hotspot resources. Specifically, this article: 1) formulates the user interest transition as a MDP to predict the regional content popularity; 2) proposes a region partition algorithm to improve the users QoE while reducing content delivery costs, and 3) conducts collaboratively cross-region content replication to offload the original CDN server. Moreover, we design an auction-based algorithm SAM to efficiently allocate the caching resource in a collaborative caching system, in which multiple CPs compete for the caching storage provided by one crowdsourced CDN provider. Our evaluation confirms the superiority of our designs. Finally, we outline some potential improvements and open issues to the crowdsourced video CDNs which are meaningful for future designs: 1) Considering the user location is time-varying under today's mobile network, the user mobility could be used for the content replication, to make the service provided by edge hotspots more stable and efficient; 2) More end-to-end QoS information (e.g., RTT or bandwidth) could be collected from pairs of content hotspots, which can be used for fine-grained user request redirection.

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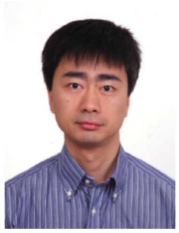

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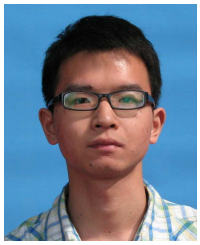

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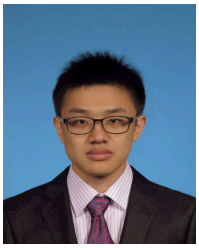

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